Business Statistics

Populations and samples
Flash Cards

Datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Population of interest
Take a sample or collect for the population?

- Business decision: Should the registrar revise advising resources for the introductory statistics course?
- Research question: What is the average final exam score for students in the introductory statistics course last semester?
- Population of interest: All students enrolled in the introductory statistics course last semester
- Required Data: Final exam scores for each student in the course
- Is it feasible and practical to collect data for the entire population? Yes
- Will measures be population parameters or sample statistics?
  population parameters

Population of interest
Take a sample or collect for the population

- Business decision: Should city planners redesign downtown traffic patterns?
- Research question: What is the average commute time for U.S. workers living in urban areas?
- Population of interest: All U.S. workers living in urban areas
- Required Data: Individual commute times, location, and work schedules.
- Is it feasible and practical to collect data for the entire population? No
- Will measures be population parameters or sample statistics?

  sample statistics

Population of interest
Take a sample or collect for the population

- Business decision: Should a small company adjust the call center weekend staffing levels?
- Research question: What is the average number of calls handled per agent during weekend shifts last month?
- Population of interest: All weekend call-center agents employed last month
- Required Data: Number of calls handled per agent during weekend shifts.
- Is it feasible and practical to collect data for the entire population? Yes
- Will measures be population parameters or sample statistics?

  population parameters

Population of interest
Take a sample or collect for the population

- Business decision: Should streaming services invest more in original content?
- Research question: What percentage of streaming subscribers cancel their service within the first year?
- Population of interest: All streaming service subscribers.
- Required Data: Subscription start dates, cancellation dates, and account status.
- Is it feasible and practical to collect data for the entire population? No
- Will measures be population parameters or sample statistics?

sample statistics